

# Common Sector Relabeling as an Unconditional Identity

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## 1 The assertion

For a blocking period  $p$ , the physical index set of the blocked tensor is canonically identified with words of length  $p$ :

$$\text{Fin}(d^p) \simeq (\text{Fin } p \rightarrow \text{Fin } d).$$

The cyclic-sector construction uses the canonical multi-index relabeling block-wise. Write  $X \sim_{\text{MPV}} Y$  when two tensors generate the same matrix product vector family at every length and every word. Write  $R(A_k^{[p]})$  for the length- $p$  blocked tensor  $A_k^{[p]}$  read through this canonical word relabeling.

The mathematical assertion is that assembling the weighted block-diagonal tensor before this relabeling gives the same MPV family as assembling after the relabeling:

$$\bigoplus_k \mu_k^p \cdot A_k^{[p]} \sim_{\text{MPV}} \bigoplus_k \mu_k^p \cdot R(A_k^{[p]}).$$

Here  $A_k^{[p]}$  denotes the length- $p$  blocking of the  $k$ -th primitive block.

## 2 Relation to the paper

The paper arXiv:1606.00608 does not state this as a separate hypothesis. In the source argument the blocked-word identification is an automatic consequence of the canonical blocking construction: blocking a tensor and then reading the result through the explicit multi-index grouping of  $\{1, \dots, d\}^p$  is definitionally the same operation. The paper therefore treats the equality as transparent and does not name it.

## 3 Status

No deviation from the source paper is present. The relabeling assertion is the identity between two descriptions of the same blocked-word coordinates. The

formalization uses the unconditional nonzero-part identity for a common cyclic-sector family directly, without introducing a separate hypothesis.<sup>1</sup>

## 4 Verdict

The relabeling identity is therefore a formal expression of the paper’s implicit blocked-index identification, not an additional mathematical assumption.

## References

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<sup>1</sup>The family-level statement is `CommonBlockedCyclicSectorFamily.reindexed'nonzero'part`; the common primitive-block theorem applies it directly on both sides. The declarations are in `TNLean/MPS/CanonicalForm/SectorComparison/CommonBlockedCyclicSectorFamily.lean` and `TNLean/MPS/CanonicalForm/SectorComparison/CommonSectorTransport.lean`.